Remaining schematic work

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9 active issues. C21: 680 / 340 / 240 pF; C22: 680 / 320 / 240 pF at 1 / 10 / 100 kHz. Meter mode was C/Auto or unsure; D was not reported. Review provisional values by circuit function before selecting any further measurements. Q3 stays fitted. Completed steps are not new bench requests.

Drawing / connectivity	Remaining data	ERC
0 fitted open pads; 0 open PIC pads; 0 dangling wire ends. 82 native/model net partitions match.	44 ceramic values; LED1 footprint blank. TP11/TP17 hidden-layer limit retained; unsupported R33 withdrawn.	1 retained U6/U6A output conflict. Six documented power-source flags; R5 supplies U2/C7 from RF rail.

Audit boundary

Native wire geometry, library definitions, all catalog pad assignments and existing evidence were checked. Zero open pads does not establish hidden copper or functional behavior. U6A is DNP; its regulator symbol and visible conflict remain by user choice. B59 supports the RA0-TP6 net merge. No new NC markers or ERC exclusions.

Order of work

Recover critical component values: All 44 unknown ceramics classified: 12 supply starting values, four conditional crystal/feedback candidates and 28 network-constrained values. Review the function-based candidate table. No new bench batch now; prioritize shared filter parameters and antenna behavior over 44 individual readings.

Finish interface and package mapping: LED colour branches and J5 board-contact/output mapping are established. LED footprint, connector fit and external motor harness orientation remain reproduction qualifications. Obtain a matching LED vendor drawing or pad geometry; qualify J5 and the other packages before laying out a replacement PCB.

Resolve hidden paths and device behavior: RF feed/output wiring, switch contacts and front-bank ANT2 routing are supported. Candidate device roles remain unresolved. TP11/TP17 routes are complete as drawn, with hidden-layer uncertainty retained as a reconstruction limitation. Scope work and component isolation are deferred by owner. The Q1 low-resistance contradiction is cleared by B84/B85; no further resistance repeat. Continue values and package geometry while candidate device roles remain qualified. Components stay fitted and waveform work is deferred.

Recover critical component values

V01 - Capacitor values and passive ratings

Established: C5 is 470uF/16V; L1/L2 reported 1.4/2.2uH. Marked resistor values are recorded. B86-B91: C21=680/340/240 pF; C22=680/320/240 pF at 1/10/100 kHz in C/Auto or unsure mode, with D absent. Function review covers all 44 unknown ceramics: 12 supply starting values, four conditional candidates and 28 network-constrained values.

Remaining: 44 original fitted ceramic values remain unconfirmed. Twelve supply replacement values are proposed separately. Crystal loading, receiver response and switched antenna values are not uniquely fixed by footprint or carrier frequency. L1/L2 test conditions and ratings remain unspecified.

Next: Use docs/CAPACITOR_VALUE_REVIEW.html to work from function and primary reference circuits. No broad sweep or repeat Cp/D batch requested. Supply values are provisional replacement choices, not recovered originals. RF and receiver values require linked network constraints; scope and component isolation remain deferred.

Close when: Chosen reconstruction values have a stated basis and appropriate validation; any unverified original nominal, dielectric or rating remains explicitly separated from a provisional equivalent.

All unrecovered ceramic values

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

Finish interface and package mapping

M01 - LED footprint and mechanical qualification

Established: LED1 is the only unassigned footprint; R30/PIC23 lights red and R31/PIC22 lights green. C5 diameter/height, S1 body/actuator size, R8 and C11 sizes are already measured. J5 rear placement and both board contacts are mapped to U5 outputs from front/rear photos.

Remaining: LED1 land pattern/manufacturer numbering and internal pairing of its externally grounded cathode pads remain unverified. C5 pitch and S1 pad fit/3D heights are reproduction qualifications, not unknown schematic wires. J5 stock-footprint pad order/pitch and external motor harness colours remain unverified.

Next: Obtain LED underside/pad geometry or a matching vendor drawing before selecting a footprint. For later PCB reproduction check C5 lead pitch and S1 pad fit; use actual 16mm capacitor and 6mm switch heights. Do not repeat body dimensions. Match J5 physical A/B contacts to the chosen connector drawing before PCB layout; record harness colours when available, without repeating visible board continuity.

Close when: LED footprint has a supported pad map; remaining package-fit assumptions are documented separately from schematic connectivity.

Resolve hidden paths and device behavior

E03 - RF supply and Q8 drive

Established: C5/C6 and Q8 source/ground/outer-drain joins supported. U2/C7 supply now fed from H_RF_VDD through R5 (22 ohm): local and endpoint readings B78-B81 support this route. U2 supply = 6.3 V; direct difference to printed VDD = +1.8 V rejects old VDD tie. Q8 output-to-ANT1 accepted from visible copper and compatible B82 4.5-ohm reading with antenna fitted. Rest gate-source voltages: P=0 V, N=+6.3 V. TI lists C14R among the DBV candidate markings; absolute maximum VCC is 6.5 V, distinct from the 5.5-V recommended maximum.

Remaining: Exact U2 identity and compatibility with measured 6.3 V remain unresolved; candidate SN74LVC2G14 recommended supply maximum is 5.5 V. Q8 switching margin, timing and RF waveform validation are deferred by owner. B82 alone is not a zero-ohm proof; output route is photo-supported.

Next: Finish non-scope schematic items first. Review U2 fitted identification and candidate ratings from marking/datasheet evidence. Owner has an oscilloscope but explicitly defers waveform measurements until later.

Close when: U2 identity and voltage compatibility resolved; RF operating waveform validation recorded separately when the owner resumes scope work. Local RF supply/output drawing is corrected; unknown C6/C7 values remain V01.

E04 - Antenna and tuning-cell topology

Established: PIC-to-control resistors and local tuning midpoints are mapped. B44/B45 midpoint-to-antenna DC readings are OL. B52 C11 outer end to ANT2 = 1.4 ohm; photo-shared outer pads of C14/C29/C44/C47 now join ANT2. Rear capacitor returns remain GND. ANT1-ANT2 is about 5 ohm with antenna fitted. The mistaken 414 kohm N-B report was corrected to B-G. Q3 L-S latest 0.48 V, L-R 2.3 V; all other diode directions OL.

Remaining: The drawn PNP B=R/E=L/C=S candidate is still not confirmed by in-circuit junction readings. Capacitor values and actual switching/resonance behavior remain unknown. The front capacitor-bank antenna connection is now established.

Next: No more antenna continuity sweep. Leave Q3 fitted as requested. Agent reviews candidate device roles against the corrected topology; isolate-device identification is deferred. RF operating checks follow only after values and suitable instrumentation are addressed.

Close when: Tuning device pin roles and switched RF behavior are supported; outer ANT2 path is already resolved.

E07 - PIR: Q2 behavior and operating supply

Established: J3/U7 ordering, U6 external pads and PIC21/TP16/Q2 output routing are established. U6 pin5 is tied to VIN; pin4 is unused. B58 measured 3.3 V at J3.1 with PIR connected; logic VDD was 4.5 V.

Remaining: Q2 terminal functions, PIR raw-signal behavior and PIC input response remain unverified. The loaded PIR supply is established at 3.3 V in the reported state; transients and other states are not characterized.

Next: Local PIR connections and 3.3 V baseline are settled. Keep Q2 identity/pin-role qualification explicit. Defer raw-signal/TP16 response and supply-transient characterization to the later scope session, as requested.

Close when: Q2 signal behavior and U6 output are supported by device or operating evidence.

E08 - Receiver and Q7 operating behavior

Established: U3 input returns and R23/R42/C40, C23/R25/R48 and DNP R47/C42 paths are reconciled. VREF supplies U3 pin5; C25.1-TP6.1 measured 1.5 ohm joins the receiver filter to PIC2/RA0. Powered: VREF 0 V, TP5 0.066 V, R19 left/right 0/4.5 V. Both R19-to-Q7 joins measured 1.5 ohm. Q7 diode results: R->L 0.74 V, L->R 1.74 V, S->L 0.74 V, L->S OL. These support PNP polarity with L as base; reverse finite reading can include parallel-board paths. B71 direct QR-QL measured 0 V, consistent with no forward base-emitter bias in the latest state and the low receiver supply; stability not reported.

Remaining: Why the receiver is inactive and its enabled-state behavior remain unverified. Earlier separate 0/4.5 V readings conflict with the later direct 0 V; do not diagnose failure or assume simultaneous identical states. Exact Q7 identity and E/C selection remain qualified. Receiver gain/filter response and MCU interface roles are unverified; DC readings do not exclude pulses.

Next: No repeat Q7 diode, continuity or baseline DC readings. Preserve latest 0 V direct bias and 0.066 V receiver-supply result. Enabled-state receiver timing and filter response are deferred to the later scope session; exact Q7 identity/E-C mapping stays qualified.

Close when: Receiver supply, bias and the existing RA0/RA1/RB4 interfaces have supported operating roles; no presumed extra output wire is required.

E09 - Supply voltages and Q1 control

Established: VSYS/RF 6.3 V, printed VDD 4.5 V, PIR supply 3.3 V with PIR attached; VREF 0 V and red LED approximately 1 Hz. TP14 is on U5.8/OUTA at 1.5 ohm, with R43.2/V015; BATTERY+ stays separate. Receiver TP5 = 0.066 V (B62). R44 = 15 kohm (B83); TP14-to-Q1 presumed gate recheck = 21.4/21.7 kohm (B84/B85). The earlier 516-ohm conflict is not reproduced and no wiring change is warranted.

Remaining: Q1 fitted identity, terminal roles and powered control behavior remain unverified. The resistance recheck is compatible with R44 in the path but does not prove an intact MOSFET or the exact remaining path. Q1/motor dynamics and R3/R4/C28 sensing role remain qualified.

Next: No further Q1 resistance batch. Retain the existing wiring and candidate pin roles; powered gate/control timing belongs to the deferred scope session. Do not replace or rewire Q1 on the basis of the superseded low-resistance concern.

Close when: Actual supply levels and control behavior are recorded without merging battery, VSYS or filtered branches.

E10 - D6 breakdown and ICSP compatibility

Established: D6 anode is GND and cathode is VPP; forward drop is about 0.7V and reverse diode mode is OL. J1 pin order is resolved. Primary programming specification DS40001753B requires 8.0-9.0 V for high-voltage entry; low-voltage entry exists only when LVP is enabled.

Remaining: G3 is nonunique: MMSZ5228BS is 3.9 V; PZU2.4B is 2.4 V. Neither rating has been measured on this board. A directly connected clamp of either rating would load the required high-voltage ICSP level. The current PIC LVP configuration and actual D6 breakdown are unknown.

Next: No repeated continuity or diode-mode check. Keep D6 fitted as requested. Isolated breakdown testing is deferred. Retain the unknown breakdown/ICSP compatibility qualification; do not infer it from the G3 marking or perform a breakdown test through the PIC.

Close when: Clamp class/rating is established or explicitly bounded for reconstruction, and programming compatibility is checked.

I01 - Candidate device identities and pin roles

Established: Manufacturer candidate datasheets and pin tables exist. D2 local wiring is complete; measured R -> S -> L diode behavior supports a series pair.

Remaining: Exact fitted identity/ratings remain candidates for D1-D6, Q1-Q8/Q11, U1/U2/U6. The 4P marking alone does not identify three-pin D2; a manufacturer 4P code for a two-pin diode is incompatible.

Next: Resolve electrical risks through E03/E04/E07/E08/E09/E10. For remaining ambiguity retain the supported equivalent circuit and disclose ratings not established; no sharper-marking request or repeat D2 rail sweep. Exact manufacturer recovery is not always possible with a meter.

Close when: Each part has a supported pin/function model and adequate ratings or an explicit accepted-equivalent limitation; exact branding is not falsely claimed.

Documented reconstruction limitations

E06 - PIC14/TP11 and PIC18/TP17: hidden continuation not excluded

Disposition: Drawing resolved; physical completeness unproved

PIC14/RC3 reaches TP11; PIC18/RC7 reaches TP17. Front/rear photos show no onward surface route. Both local routes are represented and are valid endpoints.

Unproved: Additional inner-layer branches and firmware purpose remain unverified; test-only use is not asserted.

Next: None justified for E06. Do not repeat the rejected TP16-TP11 or TP16-TP17 checks.

Reopen if: A specific unassigned destination, contradictory continuity result, inner-layer image or firmware evidence identifies an onward connection that the current drawing omits.

evidence/e06_disposition_v0941.json

PCB reproduction requirements

Electrical design

Resolve the active main-board issue groups, recover critical values, and qualify candidate pin functions/ratings. TP11/TP17 hidden-layer completeness remains a documented reconstruction limitation.

Agent: Trace visible copper, review datasheets and keep the model/native files consistent.

Owner input: Antenna outer-rail continuity complete. Future measurements are selected individually after photo review; powered behavior remains unverified.

Mechanical fit

Dimensioned board perimeter, mounting holes, connector positions and lead geometry; LED land pattern.

Agent: Create a dimensioned replacement layout from supported measurements.

Owner input: Only geometry not recoverable reliably from the photos; existing body dimensions are retained.

Copper and stackup

No routed PCB exists; four layers remain a hypothesis. Continuity identifies nets, not hidden track geometry or dielectric thickness.

Agent: Design and check a replacement routing/stackup once electrical and mechanical data are sufficient.

Owner input: Layer/board thickness evidence if exact physical replication is required.

Firmware and configuration

No PIC firmware/configuration backup is present. An identical populated board needs the program and configuration, or compatible replacement firmware.

Agent: Plan a read-only backup after ICSP/D6 compatibility is established; do not erase the original.

Owner input: Programmer availability later. Protection may prevent reading the program; this is currently unknown.

Complete working assembly

Powered validation, motor/antenna/PIR integration, and daughterboard internals if that board is also being copied.

Agent: Define staged checks against the original and validate the replacement.

Owner input: Operating measurements on the original; external assemblies and their geometry if reproduced.

Scope and sources

Main-board schematic finalization. Routed PCB/layer stack, PIR daughterboard internals and firmware reconstruction remain later work, not completed. Empty-option population design is not needed to reproduce the observed DNP board.

DNP options retain all modeled local routes and candidate roles. J6.2/R36.1 terminates at an unpopulated external interface; no additional onboard rail is assumed. U6A stays a regulator symbol with its visible output conflict by explicit user choice.

All previous issue dispositions: evidence/finalization_review_v0931.json. Native geometry and uncertainty coverage: evidence/finalization_audit.json. Current register: evidence/remaining_work.json. Archived observations and old checklists: docs/history/README.md.

Power-source flags describe supply paths through existing passive components; they do not prove voltage or transistor operation. Installed KiCad 10 libraries and project symbols match the embedded pin/graphic definitions. GUI library reload has not been verified.

Reference rails

Rail	Existing modeled path
BATTERY+	Actual positive battery terminal -> Q1 single pad/native3 (candidate drain). TP14 is OUTA, separate from this rail.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.
VREF	TP104, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.
GND	Battery negative and ground copper.